MTH302 - Endsem

1. Let $V = \mathbb{R}^2$ with the standard inner product and $W = \mathbb{R}(2,3)$, (the subspace spanned by $(2,3)$). If E is the orthogonal projection of V onto W, determine the matrix of E in the standard basis. (4)

2. Let
$$M = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}, \quad A = M^{30}$$
Find a real orthogonal matrix P whose columns form an eigenbasis of A. Is A normal? Justify your answer. (4)

3. Consider the inner product space $V = \mathbb{R}[t]$ with the inner product $\langle f \mid g \rangle = \displaystyle\int_{-1}^{1} f(t)g(t)dt$. Let $P_n(t) = \{f \in V : \deg(f) \leq n\}$.
   (a) Find an orthogonal basis for $P_2(t)$
   (b) Determine the projection of $f(t) = t^4$ on $P_2(t)$ [Context: Othogonal projection]
   (c) If D is the differentiation operator on V, find D^* (6 total)

4. State True or False and give short explanations for your answers (2 + 2 + 2)
   (a) Let T be a self-adjoint operator on a finite-dimensional inner product space over a field $\mathbb{F}$. Then, $cI + iT$ is non-singular for all nonzero scalars $c \in \mathbb{F}$
   (b) For any $n \times n$ matrix A over a field $|mathbbF$, the transpose, A^t and A have the same eigenvalues and eigenvectors.

(c) There can exist linear operators T on a vector space V such that $T \neq I$ and every subspace of V is invariant under T.

5. Answer any three

(a) Let T be an invertible linear operator on a finite-dimensional complex inner product space V. Prove thatt the following statements are equivalent:

i. $T = NU$ where N is positive, U is unitary, and N commutes with U

ii. $T = c_1E_1 + \cdots + c_kE_k$ where $I = E_1 + \cdots + E_k$, $E_iE_j = 0$ for $j \neq i$, and $E_i^2 = E_i^* = E_i$. (10)

(b) Define a positive linear operator.

i. If T is a linear operator on a finite-dimensional inner product space such that T is positive and unitary, prove that $T = I$.

ii. Prove that the product of two positive linear operators is positive if and only if they commute (1 + 4 + 5)

(c) Let W b a finite-dimensional subspace of an inner product space V. Let E be the orthogonal projection of V on W

i. Prove that $\|E\alpha\| \leq \|\alpha\|$ for every $\alpha \in V$ (Context: You have to show that E is an operator first)

ii. Prove that E is a self-adjoint operator on V. (5 + 5)

(d) Let $\mathbb{F} = \mathbb{C}$. Let V be a finite dimensional vector space over $\mathbb{F}$.

i. Suppose that $E_1, \cdots, E_k$ are projections of V and that $E_1 + \cdots + E_k = I$. Prove that $E_iE_j = 0$ for $i \neq j$.

ii. Suppose $V = M_n(\mathbb{F})$ and $A \in V$ is a diagonal matrix with characteristic polynomial $\prod_{i=1}^{k}(x - c_i)^{d_i}$ where $c_i \neq c_j$ for $i \neq j$ and $d_i \geq 1$ for each i and let $W = \{B \in V : AB = BA\}$. Prove that the dimension of W is $\sum_{i=1}^{k} d_i^2$. (5 + 5)